MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 10 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Questions

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Q1. If $\triangle ABC \sim \triangle PQR$ with $\angle A = 50^\circ$ and $\angle B = 70^\circ$, then the measure of $\angle R$ is: [1]

(a) 50° (b) 70° (c) 60° (d) 80°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 60°

We know the Triangle Angle Sum Property and corresponding angles of similar triangles:

Formula / Theorem Used: $\angle A + \angle B + \angle C = 180^\circ$ and $\triangle ABC \sim \triangle PQR \implies \angle R = \angle C$

In $\triangle ABC$: $\angle C = 180^\circ - (\angle A + \angle B) = 180^\circ - (50^\circ + 70^\circ) = 60^\circ \implies \boxed{\angle R = \mathbf{60^\circ}}$.

Q2. In $\triangle ABC$, $AB = 6\sqrt{3}$ cm, $AC = 12$ cm, and $BC = 6$ cm. The angle B is: [1]

(a) 120° (b) 60° (c) 90° (d) 45°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 90°

We know the Converse of Pythagoras Theorem:

Formula / Theorem Used: If $AB^2 + BC^2 = AC^2$ in $\triangle ABC$, then the angle opposite longest side AC is 90°.

Checking squares of sides: $AB^2 = (6\sqrt{3})^2 = 36 \times 3 = 108$, $BC^2 = 6^2 = 36$, and $AC^2 = 12^2 = 144$.
Since $AB^2 + BC^2 = 108 + 36 = 144 = AC^2$, by the Converse of Pythagoras Theorem, $\boxed{\angle B = \mathbf{90^\circ}}$.

Q3. In a trapezium $ABCD$, $AB \parallel CD$ and diagonals intersect at O. If $AB = 6$ cm and $CD = 3$ cm, then the ratio $AO : OC$ is: [1]

(a) $1 : 2$ (b) $2 : 1$ (c) $4 : 1$ (d) $1 : 4$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) $2 : 1$

In $\triangle AOB$ and $\triangle COD$: $\angle OAB = \angle OCD$ (alt. int., $AB \parallel CD$) and $\angle AOB = \angle COD$ (vert. opp.) $\implies \triangle AOB \sim \triangle COD$ (AA):

Formula / Theorem Used: $\dfrac{AO}{OC} = \dfrac{AB}{CD}$

$$\frac{AO}{OC} = \frac{6}{3} = \frac{2}{1} \implies \boxed{AO : OC = \mathbf{2 : 1}}$$

Q4. In $\triangle PQR$, S and T are points on PQ and PR such that $\dfrac{PS}{SQ} = \dfrac{PT}{TR}$ and $\angle PST = \angle PRQ$. Then $\triangle PQR$ is: [1]

(a) Equilateral (b) Isosceles (c) Right-angled (d) Scalene

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) Isosceles

We know the Converse of Basic Proportionality Theorem:

Formula / Theorem Used: If $\dfrac{PS}{SQ} = \dfrac{PT}{TR}$, then $ST \parallel QR \implies \angle PST = \angle PQR$ (corresponding angles).

Combining $\angle PST = \angle PQR$ with the given $\angle PST = \angle PRQ$ gives $\angle PQR = \angle PRQ$. Sides opposite equal angles are equal ($PR = PQ$), so $\triangle PQR$ is **Isosceles**.

Q5. Corresponding sides of two similar triangles are in the ratio $2 : 3$. The ratio of their perimeters is: [1]

(a) $4 : 9$ (b) $2 : 3$ (c) $3 : 2$ (d) $\sqrt{2} : \sqrt{3}$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) $2 : 3$

We know the perimeter ratio theorem for similar triangles:

Formula / Theorem Used: The ratio of the perimeters of two similar triangles equals the ratio of their corresponding sides.

Since corresponding sides are in the ratio $2 : 3$, the ratio of their perimeters is directly $\boxed{\mathbf{2 : 3}}$.

Q6. **Assertion (A):** The shadow of a vertical pole of height 6 m is 4 m. At the same time, the shadow of a tower is 28 m. The height of the tower is 42 m.
Reason (R): At any given instant, the sun's elevation angle is constant, making vertical objects and their shadows form similar triangles. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(a) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Because sun elevation is constant at the same instant, AA similarity gives:

Formula / Theorem Used: $\dfrac{\text{Tower Height}}{28} = \dfrac{6}{4} \implies \text{Tower Height} = \dfrac{6}{4} \times 28 = 6 \times 7 = 42 \text{ m}$

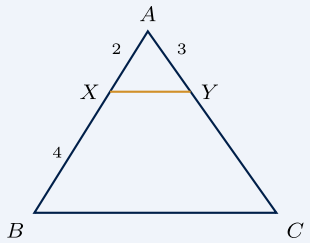
Hence both (A) and (R) are true, and (R) provides the exact mathematical reason for (A).

SECTION B: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. In the given figure, $XY \parallel BC$ in $\triangle ABC$ such that $AX = 2$ cm, $XB = 4$ cm, and $AY = 3$ cm. Find the lengths of YC and AC. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: In $\triangle ABC$, $XY \parallel BC$, $AX = 2$ cm, $XB = 4$ cm, $AY = 3$ cm.



We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: Since $XY \parallel BC$, $\dfrac{AX}{XB} = \dfrac{AY}{YC}$

Step 1: Finding Length of YC [1 Mark]

$$\frac{2}{4} = \frac{3}{YC} \implies \frac{1}{2} = \frac{3}{YC} \implies YC = 3 \times 2 = 6 \text{ cm}$$

Step 2: Finding Length of AC [1 Mark]

$$AC = AY + YC = 3 + 6 = 9 \text{ cm}$$

$$\boxed{YC = \mathbf{6} \text{ cm}, \qquad AC = \mathbf{9} \text{ cm}}$$

Q8. Prove that two triangles similar to the same triangle are similar to each other. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Two triangles $\triangle ABC$ and $\triangle PQR$ are both similar to a third triangle $\triangle XYZ$, i.e., $\triangle ABC \sim \triangle XYZ$ and $\triangle PQR \sim \triangle XYZ$.
To Prove: $\triangle ABC \sim \triangle PQR$.
We know the definition of similarity of two polygons/triangles:

Formula / Theorem Used: Two triangles are similar if their corresponding angles are equal and corresponding sides are proportional.

Step 1: Writing Equal Angles & Side Ratios [1 Mark]

From $\triangle ABC \sim \triangle XYZ$: $\angle A = \angle X$, $\angle B = \angle Y$, $\angle C = \angle Z$ and $\dfrac{AB}{XY} = \dfrac{BC}{YZ} = \dfrac{AC}{XZ} = k_1$ (constant).

From $\triangle PQR \sim \triangle XYZ$: $\angle P = \angle X$, $\angle Q = \angle Y$, $\angle R = \angle Z$ and $\dfrac{PQ}{XY} = \dfrac{QR}{YZ} = \dfrac{PR}{XZ} = k_2$ (constant).

Step 2: Establishing Transitivity [1 Mark]

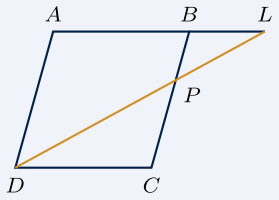
Since both triangles share equal angles with $\triangle XYZ$: $\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$.
Dividing the two side-ratio equations term-by-term:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{k_1}{k_2} \implies \boxed{\therefore \triangle ABC \sim \triangle PQR \quad \text{[Hence Proved]}}$$

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. In the given figure, $ABCD$ is a parallelogram. A line drawn through vertex D intersects side BC at P and ray AB produced at L. Prove mathematically that: $\triangle DPC \sim \triangle LPB$ and $\dfrac{DP}{PL} = \dfrac{DC}{BL}$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: Parallelogram $ABCD$ ($DC \parallel AB$, so $DC \parallel AL$). Line DPL intersects BC at P and AB produced at L.
To Prove: (i) $\triangle DPC \sim \triangle LPB$, (ii) $\dfrac{DP}{PL} = \dfrac{DC}{BL}$.
We know the parallel-lines alternate angle property and AA similarity criterion:

Formula / Theorem Used: When $DC \parallel AL$, alternate interior angles are equal; two equal angles give $\triangle DPC \sim \triangle LPB$ (AA).

Step 1: Proving $\triangle DPC \sim \triangle LPB$ [2 Marks]

In $\triangle DPC$ and $\triangle LPB$:

1. $\angle DPC = \angle LPB$ (Vertically opposite angles)
2. $\angle PDC = \angle PLB$ (Alternate interior angles, since $DC \parallel AL$, transversal DL)

By the **AA similarity criterion**, $\triangle DPC \sim \triangle LPB$ ($D \leftrightarrow L, P \leftrightarrow P, C \leftrightarrow B$).

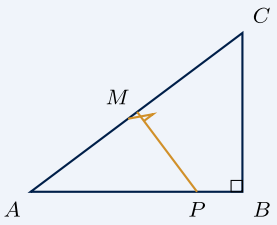
Step 2: Deducing Side Ratio [1 Mark]

Since corresponding sides of similar triangles are proportional:

$$\frac{DP}{LP} = \frac{DC}{LB} \implies \boxed{\frac{DP}{PL} = \frac{DC}{BL} \quad \text{[Hence Proved]}}$$

Q10. In the given figure, $\triangle ABC$ and $\triangle AMP$ are two right triangles, right-angled at B and M respectively. Prove mathematically that: (i) $\triangle ABC \sim \triangle AMP$, (ii) $\dfrac{CA}{PA} = \dfrac{BC}{MP}$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: In $\triangle ABC$ and $\triangle AMP$, $\angle ABC = 90°$ and $\angle AMP = 90°$, sharing vertex A.

To Prove: (i) $\triangle ABC \sim \triangle AMP$, (ii) $\dfrac{CA}{PA} = \dfrac{BC}{MP}$.

We know the AA Similarity Criterion:

Formula / Theorem Used: If two triangles have two pairs of equal corresponding angles, they are similar and corresponding sides are proportional.

Step 1: Proving $\triangle ABC \sim \triangle AMP$ [2 Marks]

In $\triangle ABC$ and $\triangle AMP$:

1. $\angle CAB = \angle PAM$ (Common angle $\angle A$)
2. $\angle ABC = \angle AMP = 90°$ (Given right angles)

By the **AA similarity criterion**, $\boxed{\triangle ABC \sim \triangle AMP}$.

Step 2: Deducing Side Proportionality [1 Mark]

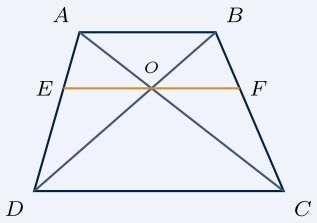
Since corresponding sides of similar triangles $\triangle ABC$ and $\triangle AMP$ are in equal ratio:

$$\boxed{\frac{CA}{PA} = \frac{BC}{MP} \quad \text{[Hence Proved]}}$$

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Q11. In a trapezium $ABCD$, $AB \parallel DC$ and diagonals intersect at O. A line segment EF is drawn parallel to AB passing through O meeting AD at E and BC at F. Prove that: (i) $EO = OF$, (ii) if $AB = a$ and $CD = b$, $EF = \dfrac{2ab}{a+b}$. **[5]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Given: Trapezium $ABCD$ with $AB \parallel DC$, $EF \parallel AB$ through O ($AB \parallel EF \parallel DC$), $AB = a$, $CD = b$.
We know BPT and AA similarity criterion:

Formula / Theorem Used: Parallel cross-lines create similar triangle pairs ($\triangle DEO \sim \triangle DAB$ and $\triangle CFO \sim \triangle CBA$) sharing equal leg ratios.

Step 1: Establishing Segment Ratios via BPT & AA Similarity [2 Marks]

In $\triangle AOB$ and $\triangle COD$: $\angle OAB = \angle OCD$ (alt. int.) and $\angle AOB = \angle COD$ (vert. opp.) $\implies \triangle AOB \sim \triangle COD$ (AA):

$$\frac{AO}{OC} = \frac{BO}{OD} = \frac{AB}{CD} = \frac{a}{b}$$

In $\triangle ADC$, $EO \parallel DC \implies \dfrac{AE}{ED} = \dfrac{AO}{OC} = \dfrac{a}{b} \implies \dfrac{ED}{AD} = \dfrac{b}{a+b}$.

Similarly, in $\triangle BDC$, $OF \parallel DC \implies \dfrac{BF}{FC} = \dfrac{BO}{OD} = \dfrac{a}{b} \implies \dfrac{CF}{CB} = \dfrac{b}{a+b}$.

Step 2: Finding EO & OF ($EO = OF$) [2 Marks]

1. In $\triangle DAB$, $EO \parallel AB \implies \triangle DEO \sim \triangle DAB$ (AA) $\implies \dfrac{EO}{AB} = \dfrac{ED}{AD} = \dfrac{b}{a+b} \implies EO = \dfrac{ab}{a+b}$.
2. In $\triangle CBA$, $OF \parallel AB \implies \triangle CFO \sim \triangle CBA$ (AA) $\implies \dfrac{OF}{AB} = \dfrac{CF}{CB} = \dfrac{b}{a+b} \implies OF = \dfrac{ab}{a+b}$.

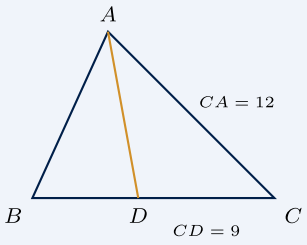
Therefore, $\boxed{EO = OF = \dfrac{ab}{a+b}}$.

Step 3: Finding Total Length EF [1 Mark]

$$EF = EO + OF = \frac{ab}{a+b} + \frac{ab}{a+b} = \boxed{\frac{2ab}{a+b}} \quad \text{[Hence Proved]}$$

Q12. In $\triangle ABC$, D is a point on BC such that $\angle ADC = \angle BAC$. Prove that $\triangle ABC \sim \triangle DAC$ and $CA^2 = CB \times CD$. If $CB = 16$ cm and $CD = 9$ cm, find CA and the perimeter of $\triangle DAC$ given that the perimeter of $\triangle ABC$ is 48 cm. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Given: Point D on BC of $\triangle ABC$ with $\angle ADC = \angle BAC$; $CB = 16$ cm, $CD = 9$ cm, and Perimeter$(\triangle ABC) = 48$ cm.
We know AA Similarity and Perimeter Ratio properties:

Formula / Theorem Used: If $\triangle ABC \sim \triangle DAC$, then
$$\frac{CB}{CA} = \frac{CA}{CD} = \frac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DAC)}.$$

Step 1: Prove $\triangle ABC \sim \triangle DAC$ & $CA^2 = CB \cdot CD$ [2 Marks]
In $\triangle ABC$ and $\triangle DAC$: (1) $\angle ACB = \angle DCA$ (common angle $\angle C$), and (2) $\angle BAC = \angle ADC$ (given).
By **AA similarity criterion**, $\triangle ABC \sim \triangle DAC$.
Corresponding sides are proportional:

$$\frac{CB}{CA} = \frac{CA}{CD} \implies \boxed{CA^2 = CB \times CD}$$

Step 2: Finding Length CA [1.5 Marks]
Substituting $CB = 16$ cm and $CD = 9$ cm:

$$CA^2 = 16 \times 9 = 144 \implies CA = \sqrt{144} = 12 \text{ cm}$$

Step 3: Finding Perimeter of $\triangle DAC$ [1.5 Marks]
Since $\triangle DAC \sim \triangle ABC$, the ratio of perimeters equals the scale ratio $\frac{CA}{CB} = \frac{12}{16} = \frac{3}{4}$:

$$\frac{\text{Perimeter}(\triangle DAC)}{48} = \frac{3}{4}$$

$$\implies \text{Perimeter}(\triangle DAC) = \frac{3}{4} \times 48 = 36 \text{ cm}$$

$$\boxed{CA = 12 \text{ cm}, \quad \text{Perimeter}(\triangle DAC) = 36 \text{ cm}}$$

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Q13. Case Study 10: Civil Engineering Trapezoidal Embankment Construction

In a highway flyover project, a cross-section of an earth embankment forms a trapezium $ABCD$ with parallel horizontal road layers $AB \parallel DC$. The top width $AB = 10$ m and bottom base $CD = 25$ m. Diagonals AC and BD intersect at drainage culvert O. A reinforcement geogrid layer EF is laid horizontally through O parallel to AB, intersecting side slopes AD at E and BC at F. [4]

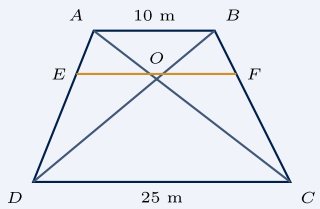
(i) **[1 Mark]** Write the ratio in which culvert O divides the diagonal AC.

(ii) **[1 Mark]** Find the ratio $\dfrac{AE}{AD}$ along the side embankment slope.

(iii) **[2 Marks]** Calculate the total width of the reinforcement geogrid layer EF.

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given Data: Trapezium $ABCD$ with $AB \parallel EF \parallel DC$, $AB = 10$ m, $CD = 25$ m; diagonals AC, BD intersect at O.
We know AA similarity for trapeziums and BPT:



Formula / Theorem Used: $\triangle AOB \sim \triangle COD \implies \dfrac{AO}{OC} = \dfrac{AB}{CD}$, $\quad EF = \dfrac{2 \cdot AB \cdot CD}{AB + CD}$

Sub-part (i): Ratio $AO : OC$ Divided by Culvert O [1 Mark]

In $\triangle AOB$ and $\triangle COD$: $\angle OAB = \angle OCD$ (alt. int., $AB \parallel CD$) and $\angle AOB = \angle COD$ (vert. opp.). Thus $\triangle AOB \sim \triangle COD$ (AA similarity):

$$\frac{AO}{OC} = \frac{AB}{CD} = \frac{10}{25} = \frac{2}{5} \implies \boxed{AO : OC = \mathbf{2 : 5}}$$

Sub-part (ii): Finding Ratio $\dfrac{AE}{AD}$ Along Side Slope [1 Mark]

In $\triangle ADC$, since $EO \parallel DC$, by BPT: $\dfrac{AE}{ED} = \dfrac{AO}{OC} = \dfrac{2}{5}$. Adding 1 after inverting (or taking total parts $2 + 5 = 7$):

$$\frac{AE}{AD} = \frac{AE}{AE + ED} = \frac{2}{2+5} = \boxed{\mathbf{\frac{2}{7}}}$$

Sub-part (iii): Total Width of Reinforcement Geogrid Layer EF [2 Marks]

Since $\dfrac{AE}{AD} = \dfrac{2}{7}$, remaining ratio $\dfrac{DE}{AD} = 1 - \dfrac{2}{7} = \dfrac{5}{7}$ (and symmetrically $\dfrac{CF}{BC} = \dfrac{5}{7}$).

In $\triangle DAB$, $EO \parallel AB \implies \triangle DEO \sim \triangle DAB$ (AA): $\dfrac{EO}{10} = \dfrac{5}{7} \implies EO = \dfrac{50}{7}$ m.

Similarly, in $\triangle CBA$, $OF \parallel AB \implies \triangle CFO \sim \triangle CBA$ (AA): $\dfrac{OF}{10} = \dfrac{5}{7} \implies OF = \dfrac{50}{7}$ m.

$$EF = EO + OF = \frac{50}{7} + \frac{50}{7} = \boxed{\mathbf{\frac{100}{7}}\ \text{m} \approx \mathbf{14.29}\ \text{m}}$$

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